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Canadian Bioinformatics Workshops



# Module 3: General Concepts in Genomics and Quality Control

Dr. Ido Hatam

CBW High-Throughput Genomics

November 4-6, 2026

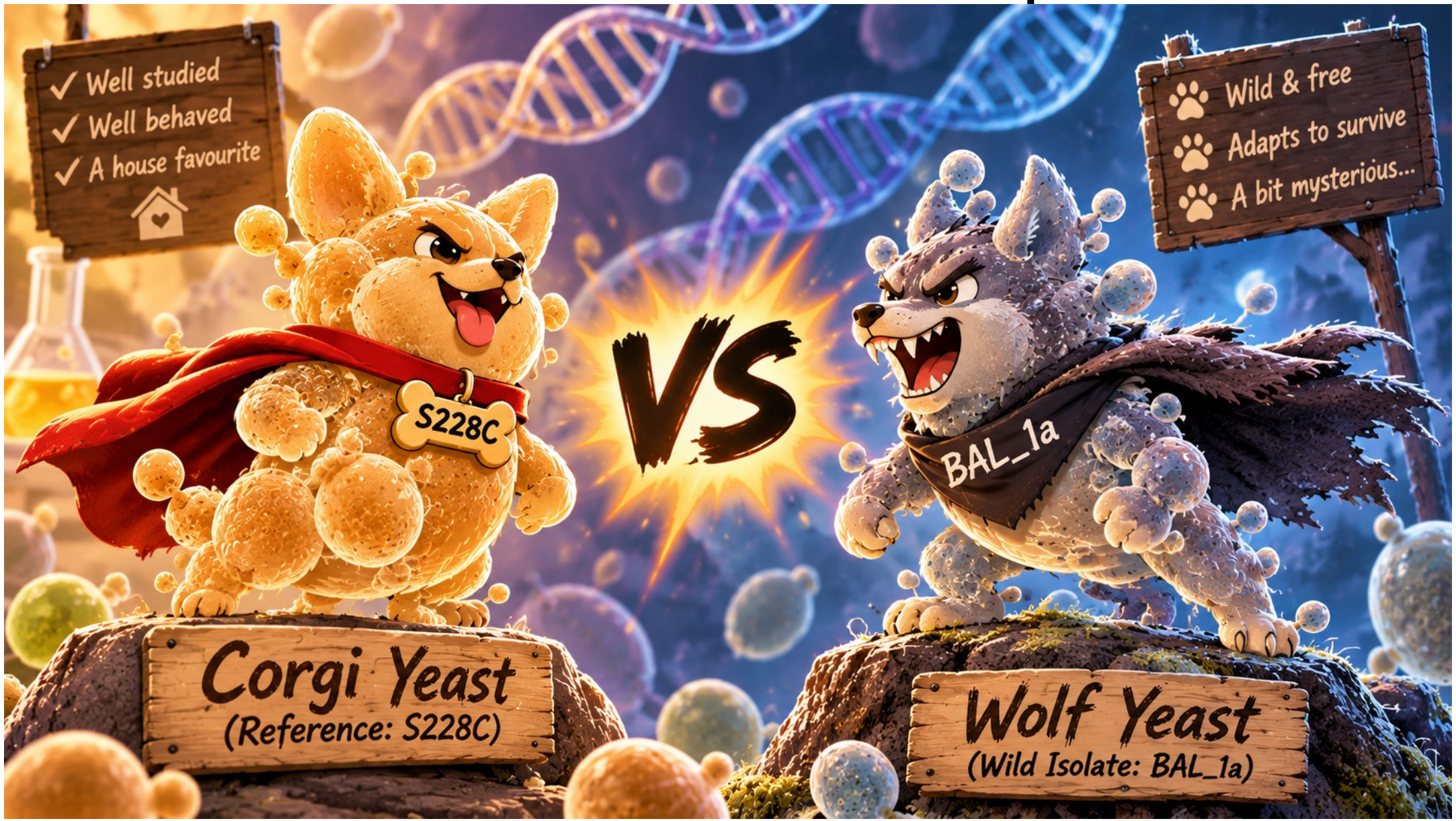
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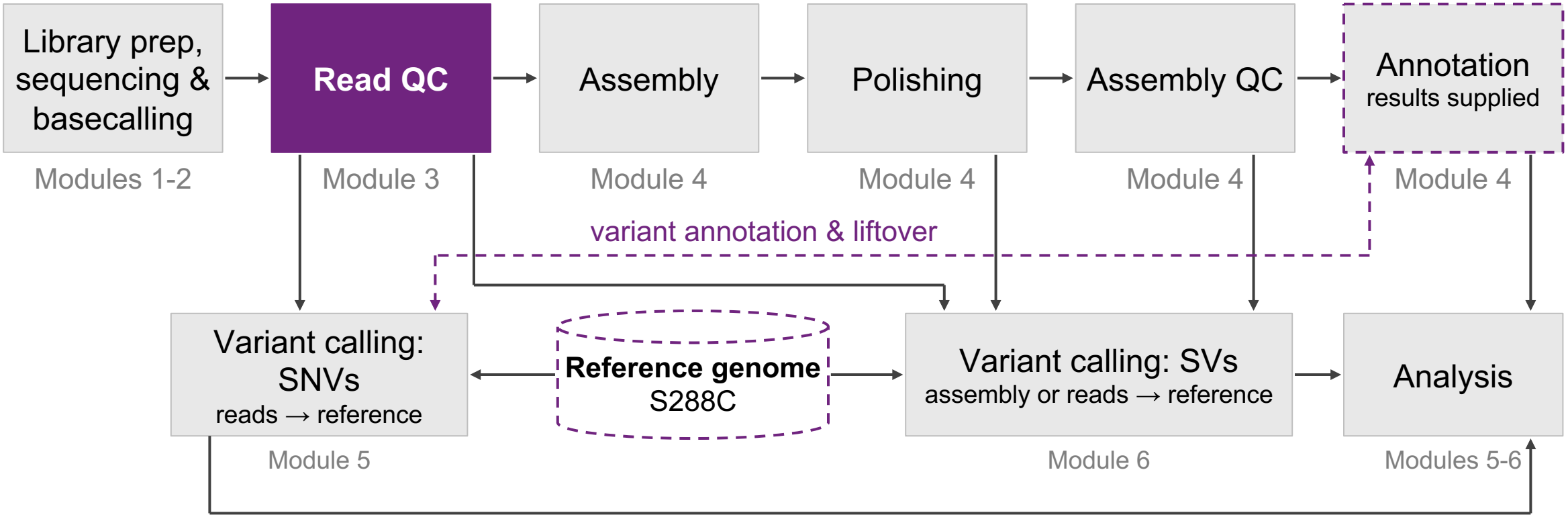
# Learning Objectives

- Describe the main sources of error in **short reads** (Illumina) and **long reads** (ONT)
- Outline the common **genomics workflow** and place each workshop module on it
- Interpret **QC metrics** for short and long reads, and judge whether the data are fit for use

# Goal for this workshop:



# How do we compare genomes? The workflow




= this module (dashed outline = supplied or reference)

# What is a genome and what are genomics

- A genome is the totality of an organism's hereditary information (DNA)
- It is more than the genes: it also includes the DNA between them
- Genomics is the scientific study of the genome: structure, function, evolution, mapping
- Comparative genomics: compare genomes to see what is shared and what differs
- Other “omes” exist (transcriptome, proteome); we start with the DNA layer

# The standard reference genome: S288C

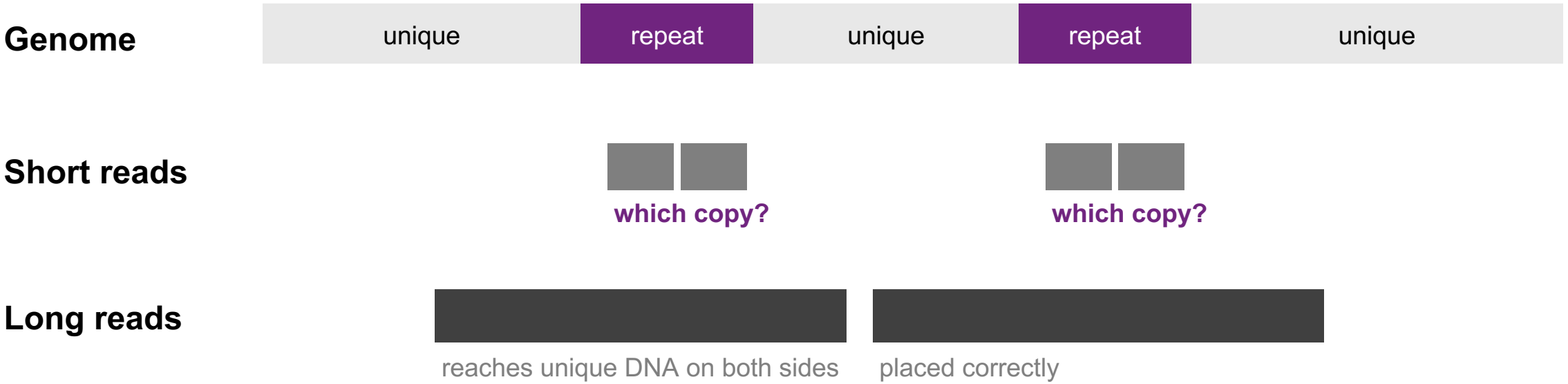
- A reference genome is an existing assembly used as a coordinate system
- For yeast the standard is S288C (release R64-1-1): the "corgi"
- Designed as a lab strain; about 88% of its genome comes from EM93, a rotting fig isolate (California, 1938)
- Reads are mapped to it, variants are described on it, genes are named on it
- Represents the whole species
- Wild isolates can differ in genes, rearrangements and repeats

# What is a T2T assembly?




- **T2T**: each chromosome is one gapless sequence, telomere to telomere
- Includes centromeres and large repeats
- Human T2T-CHM13 completed the last ~8% of GRCh38 (Nurk et al. 2022)

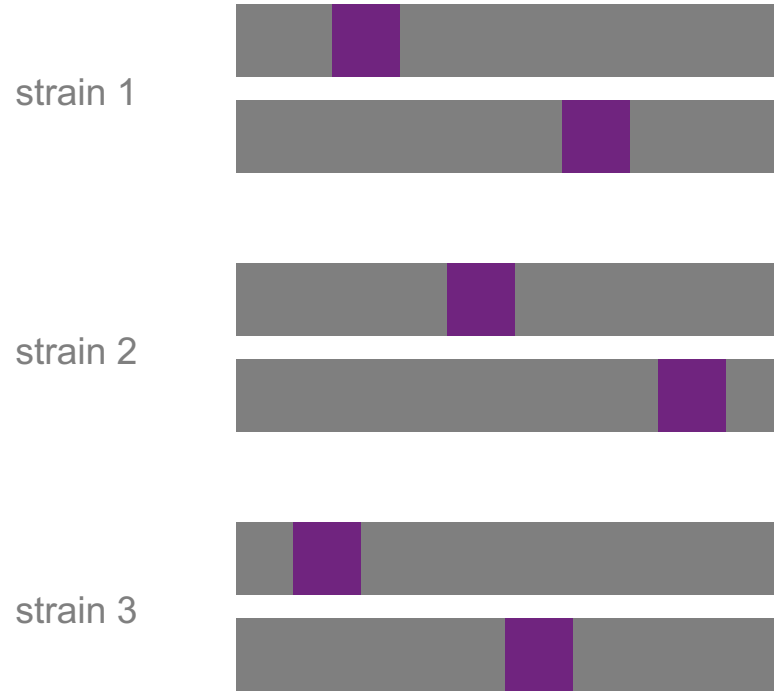
# We focus on long reads, why?




- A short read inside a repeat fits every copy: it cannot be placed.
- A long read that spans the repeat is placed with certainty.
- Long reads make T2T assembly easy(er… or possible)

# Moving away from a single linear genome

- One genome describes one individual
- **Phased** assemblies: both copies of each chromosome
- **Pangenome**: all sequence across many genomes
- **ScRAP**: 142 T2T yeast genomes, several phased (diploids, polyploids)
- Captures the species' diversity, including SVs (O'Donnell et al. 2023)




*two haplotypes per strain; purple = differences*

# S288C versus ScRAP T2T assemblies

| | S288C reference | ScRAP T2T assemblies |
|---|---|---|
| **Represents** | One lab strain, for the whole species | Each strain's own genome |
| **Chromosome ends and repeats** | A few repeat regions collapsed (e.g. rDNA) | Chromosome ends assembled; several heterozygous strains phased |
| **Strain-specific genes and SVs** | Not visible | Captured |
| **In this workshop** | The coordinate system for variants | The goal for our own assembly |

# The workflow, stage by stage (1/2)

| Stage | Goes in → comes out | Tools here | Module |
|---|---|---|---|
| **1 Library prep, sequencing & basecalling** | DNA → raw signal → FASTQ (bases + qualities) | Done for you | 1-2 |
| **2 Read QC (quality control)** | FASTQ → quality report; filtered FASTQ | sequali, filtlong, ShortRead | 3 |
| **3 Assembly** | Reads → contigs | Flye | 4 |
| **4 Polishing** | Contigs + reads → corrected contigs | medaka | 4 |

# The workflow, stage by stage (2/2)

| Stage | Goes in → comes out | Tools here | Module |
|---|---|---|---|
| **5 Assembly QC** | Assembly → contiguity, completeness | QUAST, BUSCO | 4 |
| **6 Annotation** | Assembly → where genes are, what they do | funannotate, eggNOG-mapper (supplied) | 4 |
| **7a Variant calling: SNVs** | Reads + reference → SNVs | medaka, bcftools, snpEff | 5 |
| **7b Variant calling: SVs** | Assembly or reads + reference → SVs | minimap2 + SVIM-asm (lab); Sniffles (reads) | 6 |
| **8 Analysis** | Variants + genes → biology | IGV | 5-6 |

# The wolf: BAL_1a

- A wild *Saccharomyces cerevisiae* (baker's and brewer's yeast) isolate from rotten wood, Bawangling Mountain (Hainan, China)
- ENA: “a monosporic isolate from strain BAL/HN6”
- Monosporic: grown from a single spore; a homozygous diploid
- Sequenced with Oxford Nanopore long reads: run ERR8562432
- We use all of the reads, not a subsample, from QC to variants

# Where our data come from




ScRAP: a project assembling *S. cerevisiae* strains from long reads

O'Donnell et al. 2023, *Nature Genetics*: T2T assemblies of 142 yeast strains

# The case: a wolf and a corgi

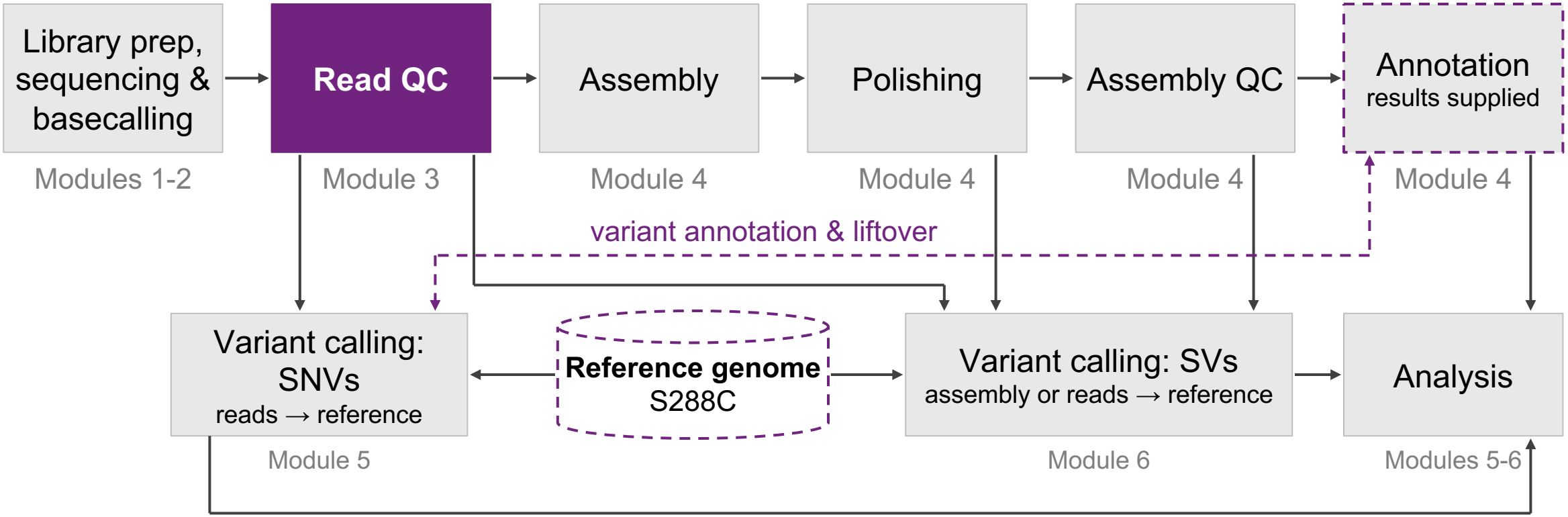
**The wolf (our isolate, BAL_1a)**
Wild; rotten wood, Hainan, China
Homozygous diploid, from one spore

**The corgi (the reference strain, S288C)**
Lab-bred; can't flock, can't mate.

We compare the two genomes: which variants separate them?
Two lines of questioning (thug life vs cushy desk job):

- Ethanol fermentation
- Stress response

# Let's begin with our workflow




= this module (dashed outline = supplied or reference)

# Our reads: ERR8562432

| | | | | |
|---|---|---|---|---|
| **MinION**<br>ONT sequencer | **R9.4.1**<br>pore type | **?**<br>reads | **?**<br>bases | **?**<br>coverage |

Coverage = total bases ÷ genome size. Illustrative example: 600 Mb ÷ 12 Mb = 50×

Library kit: SQK-LSK108 (older chemistry: expect lower quality)

The file name (BAL_1a.porechop.fq.gz) shows the reads were adapter-trimmed with Porechop

Adapter: a known synthetic sequence added in library prep; not genome

Reads, bases and coverage: you measure all three in Lab 3

# The FASTQ format: four lines per read

| | | |
|---|---|---|
| **Line 1** | **@read_001** | Read ID (starts with @) |
| **Line 2** | G A T T A C A G G C **N** T T A A C C G G A T C G A | Bases: A, C, G, T or N |
| **Line 3** | **+** | Separator |
| **Line 4** | I I I I G G D D ? ? **#** ? ? : : 5 5 0 0 + + + & & | One quality character per base |

Lines 2 and 4 always have the same length

**N**: a base the basecaller could not call

# The Phred quality score (Q)

$$Q = -10 \times \log_{10}(P)$$ where P = the probability the base call is wrong

| Q | Chance of error | Accuracy |
|---|---|---|
| 10 | 1 in 10 | 90% |
| 20 | 1 in 100 | 99% |
| 30 | 1 in 1,000 | 99.9% |
| 40 | 1 in 10,000 | 99.99% |

Higher Q = more confident call

Every +10 means 10× fewer errors

Q starts at 0; the top value depends on the platform

Illumina aims for Q30+; nanopore: about Q8-Q13 (5-15% errors). Where do ours sit? Lab 3

# Quality as one character: Phred+33

**ASCII code = Q + 33** (so Q = code − 33)

| | | | | |
|---|---|---|---|---|
| **!**<br>Q0<br>ASCII 33 | **+**<br>Q10<br>ASCII 43 | **5**<br>Q20<br>ASCII 53 | **?**<br>Q30<br>ASCII 63 | **I**<br>Q40<br>ASCII 73 |

One character per base, instead of two digits

Decode: “?” is ASCII 63, and 63 − 33 = 30, so **Q30**

Older files used +64; modern FASTQ uses +33

# N: an ambiguous base

- **N**: the basecaller could not decide between A, C, G and T
- It writes N, usually with a very low quality score (Illumina: Q2 “#”)
- Many Ns make a read suspect
- Short-read pipelines often drop reads with any N
- Not the same as **N in an assembly**: runs of N mark gaps between joined contigs

**N in a read**




A, C, G or T? Unknown

**N in an assembly**




gap of unknown sequence

# What is quality control (QC)?

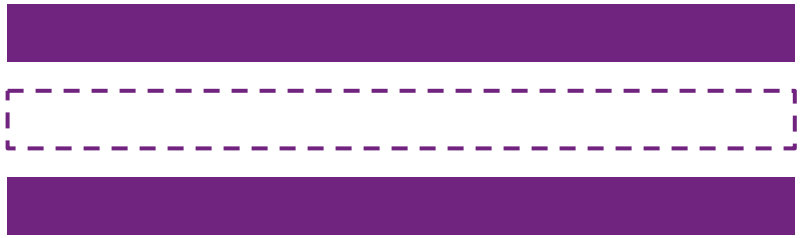
- **QC**: measuring data quality before you use it
- Garbage in, garbage out
- **Filtering** removes whole reads
- **Trimming** cuts parts of reads
- Metrics: per-position quality, mean read quality, read length distribution, read N50, Ns
- Tools: ShortRead (Illumina), sequali and filtlong (ONT)

**Trimming**



ends cut, read kept shorter

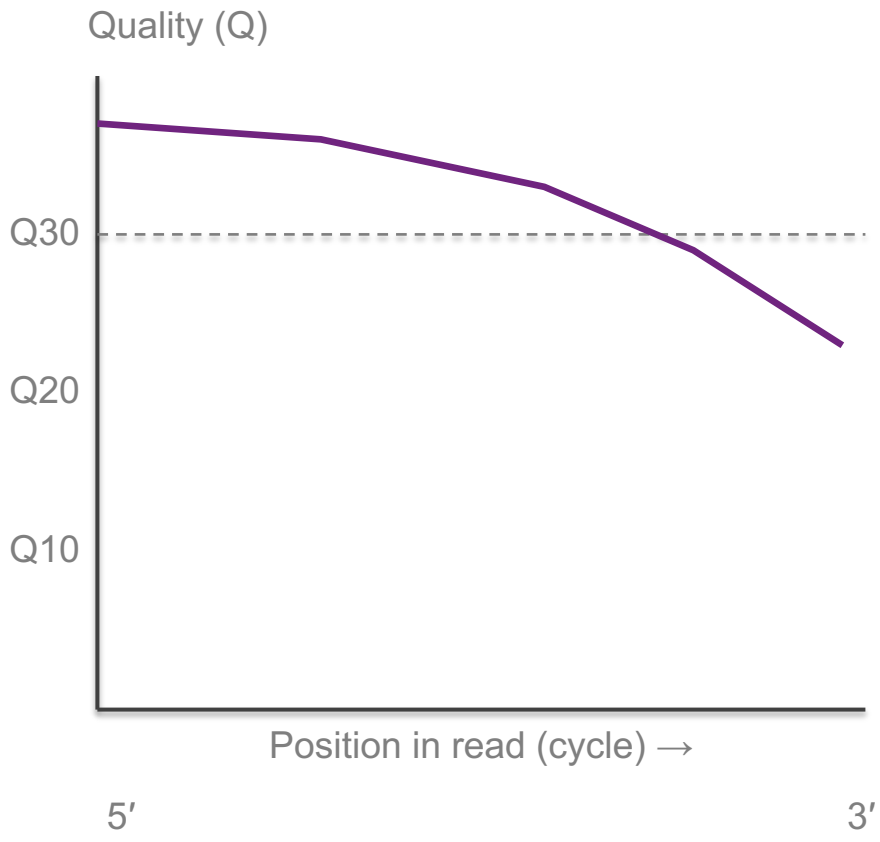
**Filtering**



whole read kept or removed

# Recap: errors in short reads (Illumina)

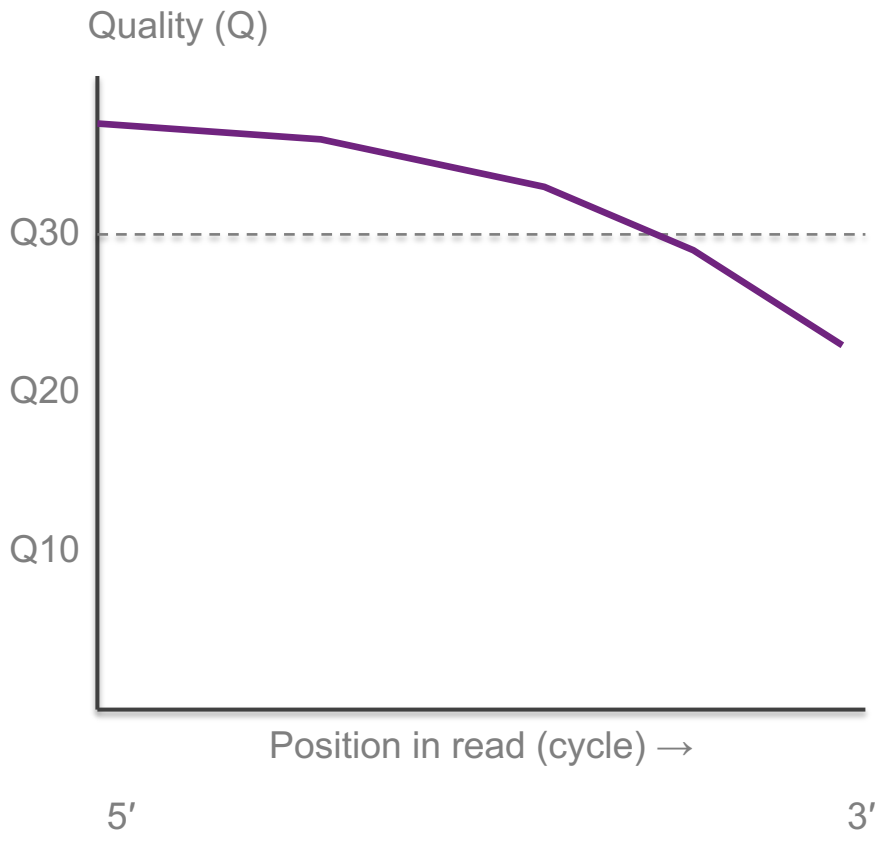
- Short reads: about 150-300 bp, accuracy above 99.9%
- Every copy in a cluster is read together, one base per cycle
- **Phasing**: some copies fall out of step (lead or lag), so the signal blurs
- Errors build up with each cycle: quality falls toward the **3′ end**
- Read 2 (R2) is usually worse than read 1 (R1)




*Schematic, not real data*

# Recap: errors in short reads (Illumina)

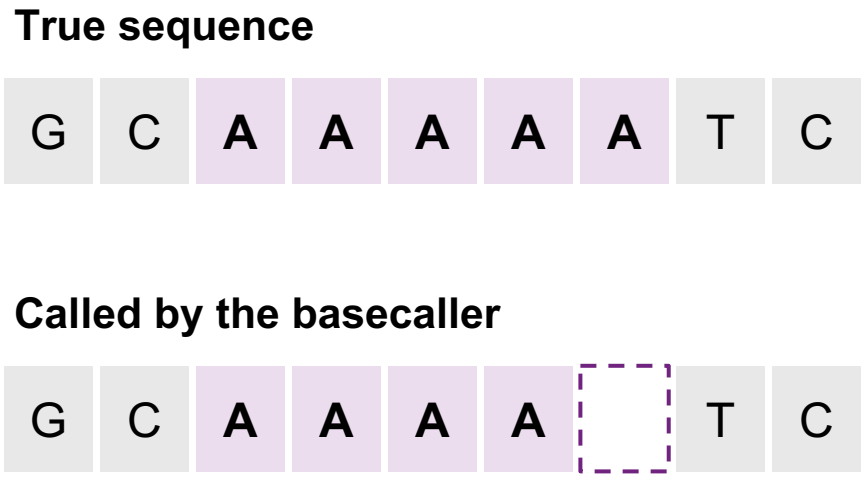
- Why do you think that R2 is worse?
- What does the error source tell you about the potential for Illumina to produce longer reads?




*Schematic, not real data*

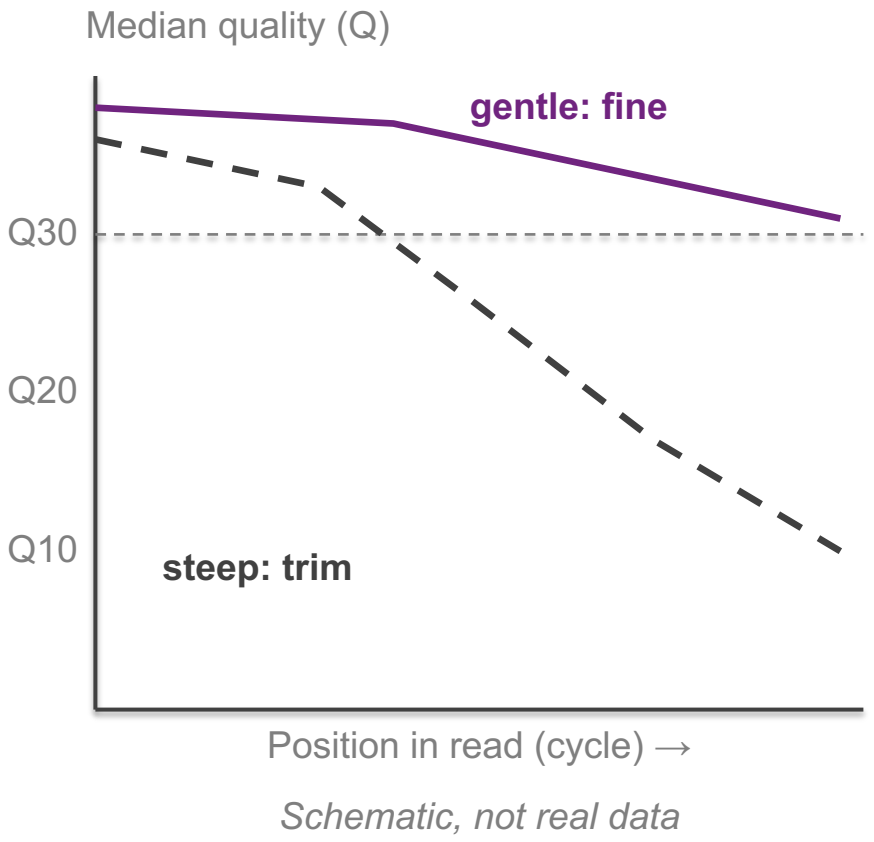
# Recap: errors in long reads (ONT)

- Current changes as one DNA strand passes through a **nanopore**
- Several bases sit in the pore at once: the signal is a blurred mix
- **Homopolymers** (runs of one base): the count is hard to read
- Errors are mostly insertions and deletions (**indels**), and partly systematic
- About 5-15% errors. No decline along the read



# Short reads: per-position quality

- **Per-position quality**: quality at each position along the read (a **cycle** is one position)
- Quality starts high and falls toward the 3′ end
- R2 is usually worse than R1
- Ask: where does the median cross **Q30**?
- A steep drop means trim the end, or a poor run




*Schematic, not real data*

# Short reads: composition, duplicates

**Base composition per cycle**

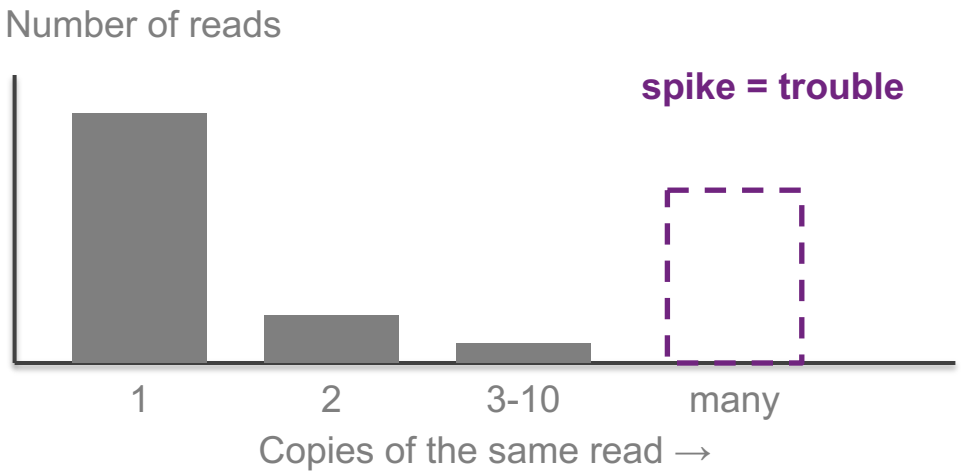



Expect flat, parallel lines; A ≈ T and G ≈ C, at levels set by the genome's GC (yeast: A, T ≈ 31%; G, C ≈ 19%)

Wobble in the first ~10 bases is common

Spikes or split lines: adapters or GC bias

**Read duplication**




Most reads should appear once

A spike of identical reads: possible PCR bias

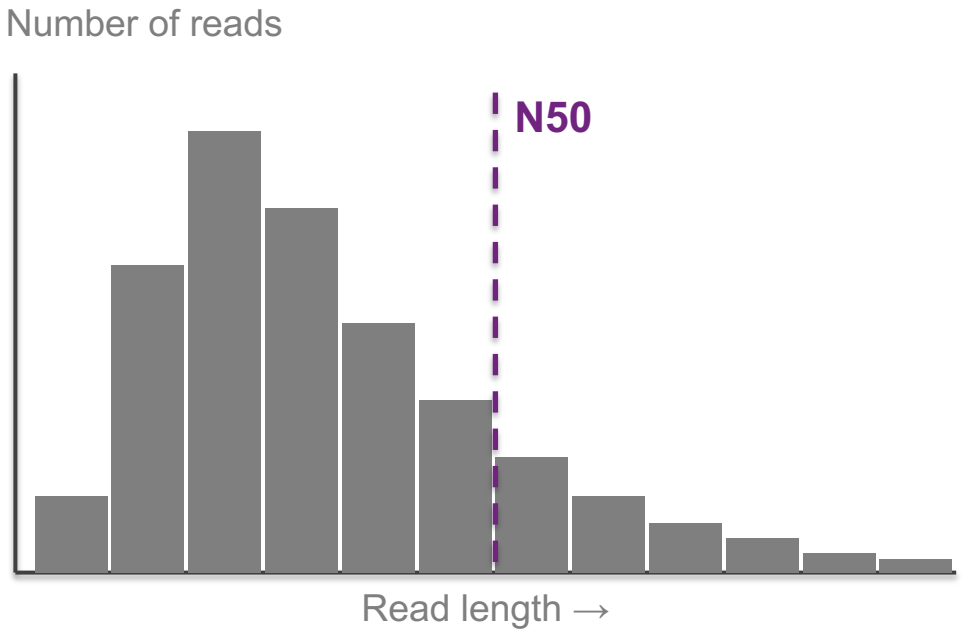
Also check: reads with **N**

# Long reads versus short reads

| | Short reads (Illumina) | Long reads (ONT) |
|---|---|---|
| **Read length** | Fixed length, 150-300 bp | A distribution: hundreds of bp to over 100 kb |
| **Quality along the read** | Falls toward the 3′ end | Roughly stable, no 3′ drop |
| **Typical median quality** | Q30 or higher | About Q8-Q13 (5-15% errors); about Q20 (newest ONT chemistry) |
| **Files per sample** | Two (R1 and R2) | One FASTQ |

# Read lengths and read N50

**Read length distribution (schematic)**

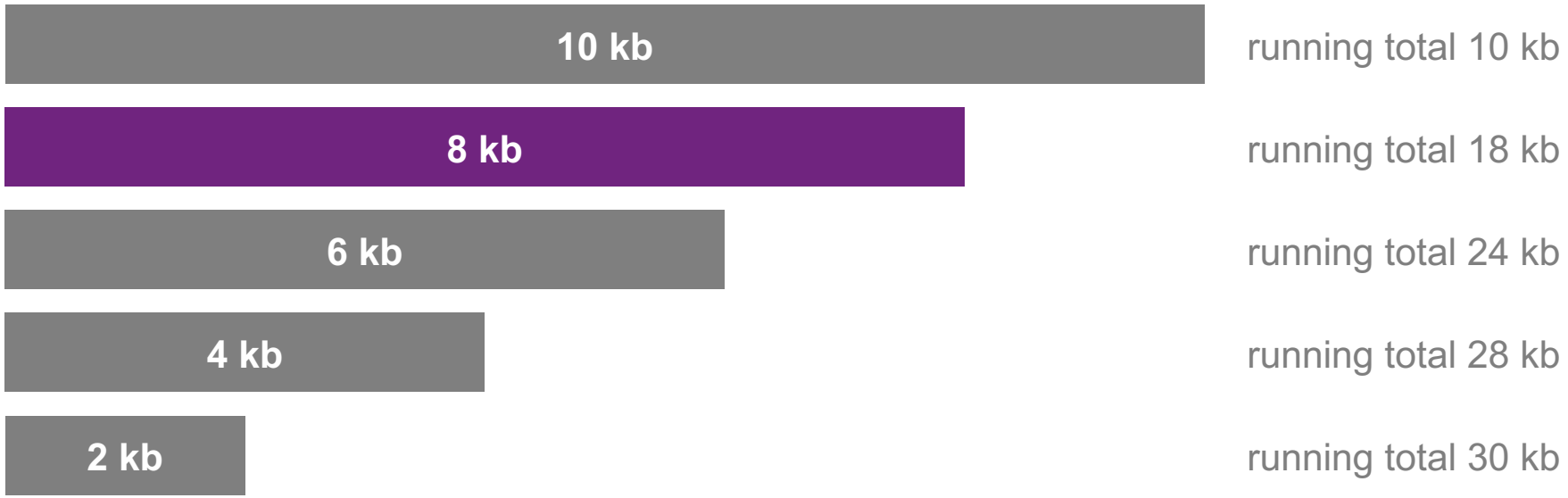



Many shorter reads, a tail of very long ones

**Read N50**
A length-weighted median: 50% of all bases are in reads at least this long

Example: read N50 = 2 kb
means half of your bases sit in reads of 2 kb or longer

# Read N50: a worked example




All reads: 30 kb. Half: **15 kb**. 10 kb is not enough; 10 + 8 = 18 kb passes it

**Read N50 = 8 kb** (the mean is 6 kb)

# Mean read quality

**A read with 10 bases**




Average the Q values: Q20 (looks good)

Average the error probabilities:
(0.001 + 0.1) ÷ 2 = 0.0505
→ about **Q13**

**The poor bases dominate, so use this version**

**Your turn in Lab 3**
Which average does sequali's per-read quality use? Look at its histogram in Lab 3

# Read QC metrics: what and why

| Metric | What it shows | Why it is useful |
|---|---|---|
| **Read length distribution** | How many reads at each length | Is the library mostly long reads or short fragments? |
| **Per-position quality** | Quality at each position along the read | Finds stretches of poor quality that trimming might target |
| **Base composition** | % of A, C, G, T at each position | Flat lines expected; bumps point to adapters or bias |
| **GC content** | % of G + C per read | One peak near 38% (S288C); extra peaks suggest contamination |
| **Ns** | Share of N bases | Many Ns make a read suspect |

**sequali**, our long-read tool, reports all of these

# Are the data fit for use?

| Ask | Short reads | Long reads (ONT) |
|---|---|---|
| **Overall quality as expected?** | Median above Q30 | About Q8-Q13 (5-15% errors) |
| **Quality steady along the read?** | Small 3′ drop; steep drop: trim | Should be stable |
| **Lengths as expected?** | One fixed length | Broad spread; check read N50 |
| **Few Ns?** | Ideally none | Few; many make a read suspect |
| **Enough data?** | Coverage is enough | Coverage is enough (aim for 30-50×) |

**Fit for use** = as expected for the platform, stable, and enough data

# Trim or filter?

| | Short reads | Long reads |
|---|---|---|
| **Where errors are** | Build up at the 3′ end | Scattered along the read; vary from read to read |
| **Trimming** | Trim adapters and 3′ ends below ~Q20; then filter short reads and reads with N | Nuanced: no one-size-fits-all recipe |
| **Our approach** | Trim, then filter | **Filter, don't trim** (a rule of thumb) |

Trimming shortens long reads, and length is their strength

# Tools you may meet with short reads: QC and trimming

| Tool | What it does |
|---|---|
| FastQC | QC report per FASTQ file |
| fastp | QC plus adapter and quality trimming in one step |
| Cutadapt | Adapter (and quality) trimming |

**Not covered in this workshop (our bonus uses ShortRead)**

# Filtering trade-off: filtlong

```
filtlong --min_length 1000 --keep_percent 90 reads.fastq.gz | gzip > filtered.fastq.gz
```

| Setting | What it does |
|---|---|
| --min_length 1000 | Drops reads shorter than 1,000 bp |
| --keep_percent 90 | Keeps the best 90% of bases, ranking reads by length and quality |

Gain: less junk in the assembly

Cost: fewer bases, lower coverage

In the lab: **not run**. Work out what it would remove, then assemble the raw reads.

# Summary

- Short reads lose quality toward the 3′ end
- ONT errors are mostly indels, often at homopolymers
- QC is stage 2 of the workflow
- FASTQ: four lines per read; Q = −10 $\log_{10}$(P error)
- Check lengths, read N50, per-position and mean read quality, GC, Ns
- Judge data against expectations; long reads: filter, don't trim

# Lab 3 preview: what you will do

| | | | |
|---|---|---|---|
| **1** Set up folders (nano, .sh) | **2** Download reads and reference | **3** Check the tools | **4** Run sequali with sbatch |
| **5** Copy the report with scp | **6** Think through filtlong | **7** Submit the Flye assembly, then lunch | **Bonus** ShortRead on Illumina data |

**Questions are about reading results, not writing code**

# Coffee Break & Networking Session

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